

CMPE 492

Cold Diffusion and Adversarial Robustness

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1. INTRODUCTION

1.1. Presentation Video

The video in the link below includes the presentation of the project.

[Video](#)

1.2. Project Code

The repository in the link below includes the implementation codes of the project and some sample images.

[Repository](#)

1.3. Broad Impact

It has been a big problem that neural networks can be deceived by adversarial attacks. Those attacks are typically carried out by adding imperceptible perturbations into the inputs of the model, which leads to incorrect predictions. To address this problem, many simple and effective methods have been created to make neural networks more robust against these attacks.

One of the most frequently used methods is adversarial training. With adversarial training, where we mix both original and adversarial samples into the training set before training the model. Then, the model is trained with this mixed data. Although effectiveness of improving the model's robustness is observed, generalization ability of this method is low and success rate may decrease against previously unseen data.

In recent years, generative models such as Generative Adversarial Networks (GANs)

[1], Diffusion Probabilistic Models [2] have emerged as a promising approach to improve the resilience of machine learning models against adversarial attacks [3]. Generative models are effective in a broader range of attack scenarios, especially if the generative model can accurately capture the data distribution. They can potentially defend against a variety of attack types without needing to be trained explicitly on adversarial examples. By adding random noise, also known as the forward process, and then denoising through a reverse process, we can create new images that fit the distribution of the existing dataset. This forward and reverse processes can also be a way to defend our models against attacks. When a random noise is applied to the image, the reversing of noise can help to not only restore the image but also counteract the adversarial attack.

In this study, we make use of a model that has a different approach to forward processing among generative models, "Cold Diffusion" [4]. Unlike random noise, cold diffusion introduces deterministic noise. Our goal is to utilize this method to achieve adversarial robustness.

Subsequently, we adjust the model's learning approach by directly applying adversarial attacks in its forward process. This means we train the model to actively recognize and neutralize these attacks as they occur instead of a degradation method. By doing so, we aim to directly improve the model's ability to defend adversarial attacks, making it more robust and secure against potential threats.

1.4. Ethical Considerations

In this research, all data we have used is taken from public sources. The research does not contain personal information or any violations.

2. PROJECT DEFINITION AND PLANNING

2.1. Project Definition

In this project, we explore whether we can provide adversarial robustness through a generative model that employs a variety of degradation methods. Generative models are based on the concept of random noise removal. It has been demonstrated with Cold Diffusion that image generation can be done without the need of reliance to the concept of randomness [4].

Using deterministic forwarding methods, we add a specific type of noise to the images, meaning exactly what kind of changes will be made in each step are determined, unlike random noise. Numerous approaches are demonstrated in the Cold Diffusion [4], like blur, animorph, mask, pixelate, and snow. They all contain different methods and calculations. Among these methods we chose the official implementation of the snowification transform from ImageNet-C[5]. The main reason we chose snowification is that the pixel information on the image is not completely lost after the forward process. Thus, after applying the reverse process, we obtain an image very close to the original image. So, in a way, we correct the lost pixel information step by step.

Then, we divide the project into two setups. The first involves applying the degradation method, specifically "snowification", to images and reverting this effect through a generative model, UNet. The second setup evolves the concept by substituting the snowification degradation with direct adversarial attack. In this setup, at each time step, we apply an adversarial attack, increasing the severity of the attack (epsilon value) as time progresses. This approach challenges the UNet model to learn how to revert these attacks, similar to its capability with snowification, thereby teaching the model to revert the attacks directly.

2.1.1. Snowification-based Degradation and Restoration

For the first setup, as observed in DENSEPURE[6] and DiffDefense[3], when we try to restore the noise-added image by reverse process, that is, to clear the noise step by step, we also aim to correct the effects of the adversarial attack. By doing so, we hope to enhance the resilience of image processing systems against such attacks, ensuring their reliability and integrity in various applications. Before the attack, we pass an image through the trained ResNet classifier. Then we attack and ensure that it is misclassified. Later, we correct this attack with the cold diffusion method and ensure that it is classified correctly again.

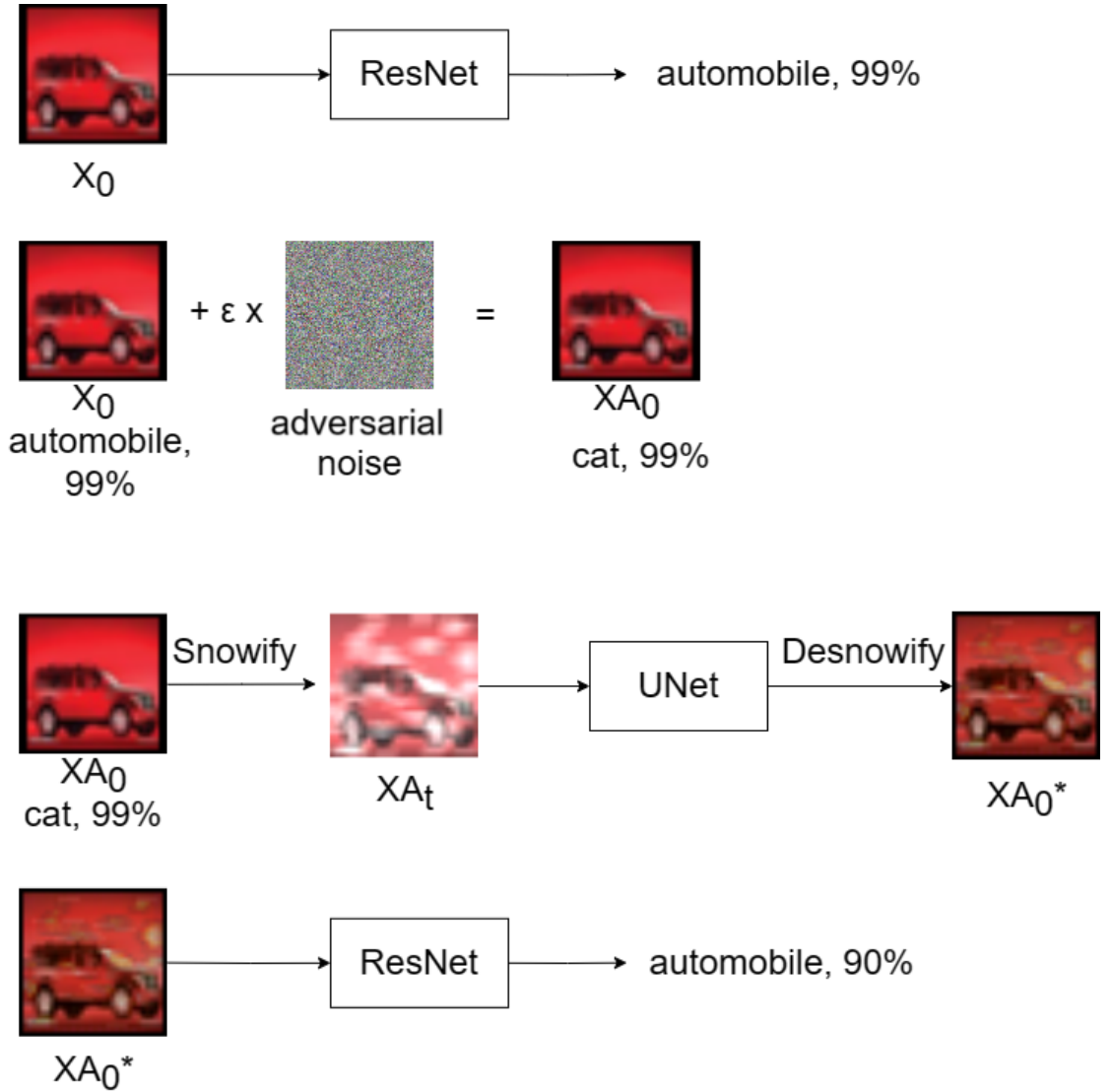


Figure 2.1. Planned example scenario

2.1.2. Adversarial Attack-based Degradation and Restoration

For the second setup, we directly integrate an adversarial attack into the degradation process instead of solely relying on the snowification method. This approach directly exposes the generative model UNet to adversarial attack's itself. By increasing the intensity of the attack step by step over time, by incrementing the epsilon value, we simulate a forward process. Then, the model is trained to reverse these attacks, learning to correct the adversarial distortions, thereby restoring the original state of the images.

Given an image $x_0 \in \mathbb{R}^N$, consider the *degradation* of x_0 by operator D with severity t , denoted $x_t = D(x_0, t)$. The output distribution $D(x_0, t)$ of the degradation should vary continuously in t , and the operator should satisfy

$$D(x_0, 0) = x_0.$$

In the standard diffusion framework, D adds Gaussian noise with variance proportional to t . In our generalized formulation, we choose D to perform various other transformations such as snowification with severity that depends on t . In this setup, D will be directly a FGSM (Fast Gradient Sign Method) attack.

We also require a *restoration* operator R that (approximately) inverts D . This operator has the property that

$$R(x_t, t) \approx x_0.$$

In practice, this operator is implemented via a neural network parameterized by θ . The restoration network is trained via the minimization problem

$$\min_{\theta} \mathbb{E}_{x \sim \mathcal{X}} \|R(D(x, t), t) - x\|,$$

where $\|\cdot\|$ denotes the Euclidean norm. [4]

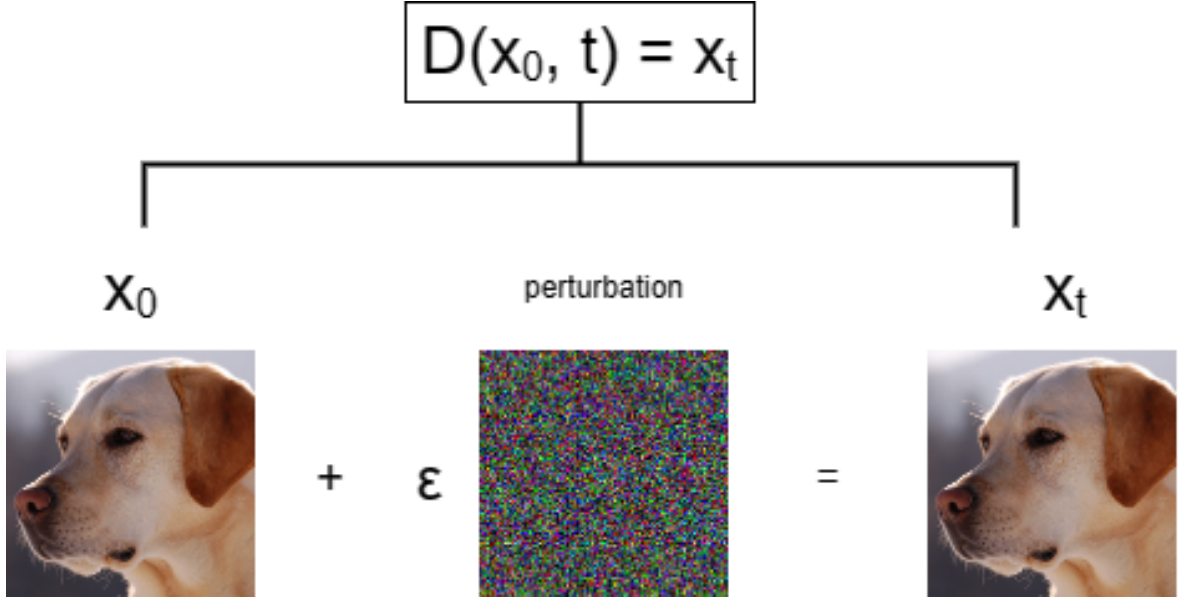


Figure 2.2. Degradation

As the timestep increases, the epsilon value, which serves as the multiplier for perturbation, increases. This leads to more noise being added to the images, with the effects of this perturbation becoming stronger as it approaches timestep t . In this context, perturbation refers to adding noise to the input image x_0 in the direction of the gradient of the loss function with respect to the input. This process is illustrated in Figure 2.2, showing the gradual degradation of the image.

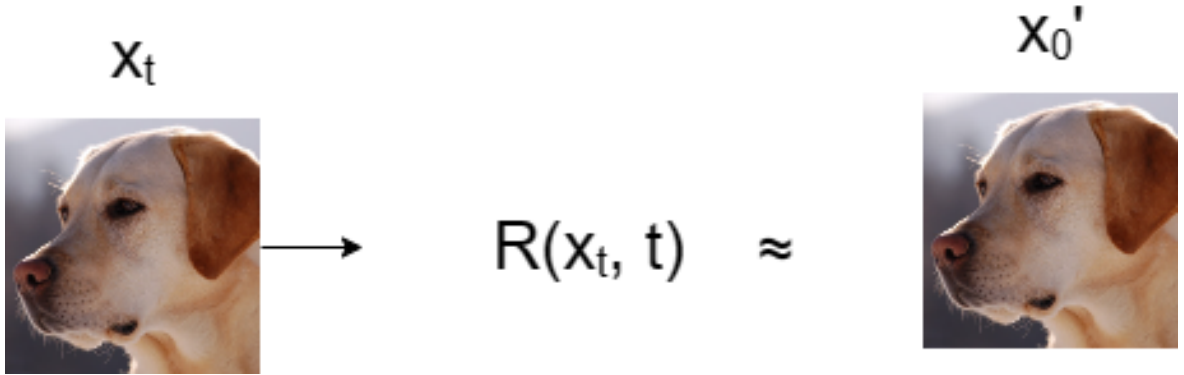


Figure 2.3. Reverse

UNet is trained to minimize the difference between the original image x_0 and the reconstructed image x_0' using the specified loss function. This process, illustrated in Figure 2.3, shows how the UNet attempts to reverse the degradation effect applied to the image x_t , producing an output that is as close as possible to the original image.

2.2. Project Planning

2.2.1. Project Time and Resource Estimation

The project spanned over a period of 9 months. Initially, the focus was on foundational research. This involved a thorough review of the existing literature on standard and cold diffusion models and a deep dive into their theoretical and mathematical foundations.

Setting up the necessary computational environment for training and testing the cold diffusion models was a key part of this early stage. We set up the [implementation](#) on Google Colab to enable synchronized teamwork. Then we created a common Github Repository and continued to develop on our own local computers.

Later, our department provided us access to HPC clusters. We used Docker images to use clusters. With high process power, our development has accelerated.

The project required high-performance computing for training and testing our models. Dataset of images, including those subjected to adversarial attacks, was essential for training and validation. We trained over 20 models, with some undergoing tens of thousands and others hundreds of thousands of epochs.

2.2.2. Success Criteria

The success of this project can be evaluated based on several key criteria:

- (i) **Development of Cold Diffusion Techniques:** Successfully creating and refining cold diffusion methods, specifically snowification, to effectively apply deterministic degradation to images. The techniques should be adaptable and efficient in various image processing scenarios.
- (ii) **Development of Adversarial Attack-based Degradation Techniques:** Effectively integrating and implementing adversarial attack-based degradation, particularly using methods like FGSM to introduce and incrementally increase the severity of adversarial perturbations over time.
- (iii) **Effectiveness Against Adversarial Attacks:** Demonstrating a significant improvement in the resilience of images against adversarial attacks post-application of both cold diffusion and direct adversarial attack-based techniques. This includes the ability to restore images to their original state or close to it after they have been attacked and subsequently processed.
- (iv) **Validation Through Comparative Analysis:** Achieving favorable outcomes when comparing our methods against existing adversarial robustness techniques. This involves showcasing better performance or distinct advantages in specific use-cases.
- (v) **Documentation and Publication:** Successfully documenting the research process, methodologies, and findings in a comprehensive manner.

2.2.3. Risk Analysis

Several risks could potentially impact the project's success:

- (i) **Technical Challenges in Model Development:** The complexity of developing new cold diffusion techniques may present unexpected challenges, particularly in fine-tuning them for specific types of adversarial attacks.
- (ii) **Adversarial Attack Complexity:** There is a risk that the adversarial attacks may evolve to be more complex than the current model's capacity to reverse, which could diminish the effectiveness of the adversarial-based degradation and restoration techniques.
- (iii) **Computational Resource Limitations:** The project's dependency on high-performance computing resources means that limitations in these resources could delay the research and development process.
- (iv) **Dataset Limitations:** The effectiveness of the models depends on the diversity and quality of the image datasets used for training and testing. Limited or biased datasets could influence the results and reduce the generalizability of our methods.

2.2.4. Team Work

Since we are a group of two, we need to work together in synchronization. We try to communicate as much as possible and save our works and their descriptions to our collaborative workspace. We are using Google Drive to store common works and reports. Also, we push our work to Github repository of the project.

3. RELATED WORK

We mostly focused on "Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise" [4]. This paper introduces cold diffusion and includes all the important details about the technique used in our project. In contrast to standard diffusion models, just add Gaussian noise, they use deterministic image transformations such as blurring, masking, resolution downsampling and snowification. This allows Cold Diffusion models to generate high-quality images without the need for noise reduction. They show accuracy of these models for creating high-quality images again. They use different metrics for evaluation such as FID, SSIM, RMSE.

"A Self-Supervised Purification Mechanism for Adversarial Samples" [7] in this study it is mentioned that adversarial training is not sufficient and its generalization capability is low. They manage to increase generalization ability by using input processing-based methods in addition to adversarial training. We examined this study in which adversarial training was used and the method was enhanced.

Another work is "DENSEPURE: UNDERSTANDING DIFFUSION MODELS FOR ADVERSARIAL ROBUSTNESS" [6]. It tries to dive into theoretical analysis of why diffusion models are thought as robust for attacks. It contains theoretical background and results that the generative model provides adversarial robustness.

The paper "DiffDefense: Defending against Adversarial Attacks via Diffusion Models" [3] discusses a new way to use Diffusion Models to keep machine learning classifiers safe from attacks that try to trick them. This method works directly with the classifiers and doesn't need any changes to them. Even though diffusion methods are usually not preferred for defense because they are slow, this research shows that DiffDefense method is strong against attacks, doesn't lose accuracy on regular data, is fast, and is easy to add to existing systems.

4. METHODOLOGY

In this section, we introduce the technical details and methods that achieve the adversarial robustness.

4.1. Forward Process

This section outlines the procedures involved in the forward process, focusing on two primary aspects: snowification and adversarial attack.

4.1.1. Snowification-based

As mentioned, we apply snowification for the forward process using the implementation in [Cold Diffusion Models](#). This is achieved through a series of image processing steps. Initially, a snow layer is created using a Gaussian distribution with parameters set depending on the desired snow density. We set snow level parameter before training and testing. Depending on the snow level, different sets of parameters are applied, such as snow threshold, motion blur radius and sigma, and brightness coefficient. As the process progresses over a number of timesteps, the snow layer’s intensity and motion effects are dynamically modified. This is done by incrementally adjusting the thresholds for snowy pixels, gradually enhancing the motion blur effect, and varying the brightness coefficients. These changes simulate the natural progression of a snowfall, from light to heavy, with increasing blur representing the movement of falling snowflakes. For motion blur, a convolution operation with a Gaussian kernel is used. Finally, the original image is combined with these snow layers.

4.1.2. Adversarial Attack-based

For the adversarial aspect, gradients for each image are generated utilizing the Fast Gradient Sign Method (FGSM) with varying epsilon values within a specified

range. This range is defined by minimum and maximum epsilon values, and the rate of adjustments is correlated with the number of timesteps. The generated gradients are then added to the image in every forward process corresponding to a given timestep. This technique introduces small but intentional perturbations to the original image, aiming to deceive machine learning models by altering the image data while keeping the modifications imperceptible to the human eye also using timestep logic in training.

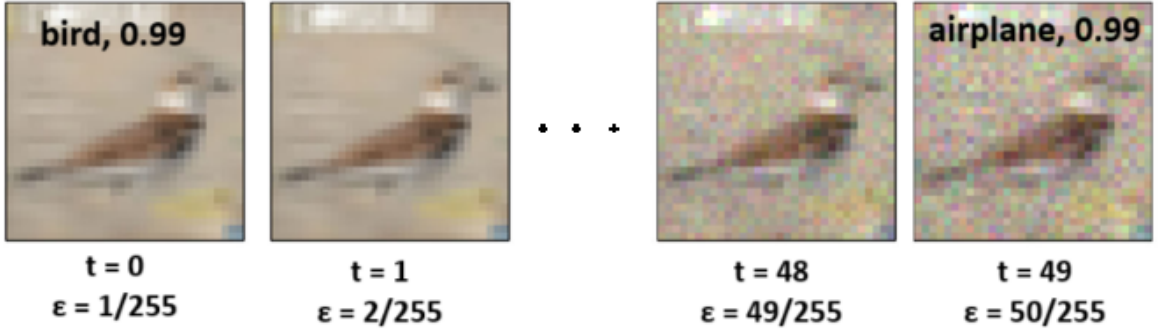


Figure 4.1. Forward process with FGSM

For example, if we set the minimum epsilon value to $1/255$, the maximum epsilon value to $50/255$, and distribute these values over 50 timesteps, then at the first timestep, an epsilon value of $1/255$ is multiplied by the perturbation and added to the image. At the second timestep, $2/255$ is multiplied by the perturbation and added, and so on, until the final timestep where $50/255$ is multiplied by the perturbation and applied. This ensures a gradual increase in the perturbation magnitude over the 50 steps.

4.2. Reverse Process

The reverse processes of generative models are based on gradually correcting the noise-added sample. At each timestep, the model predicts the noise that was introduced in the preceding step and then removes this predicted noise from the current sample, thus iteratively denoising the sample for that specific timestep. However, the algorithm stops just before the final step. This standard reverse process algorithm does not give good results for the forward process applied in cold diffusion. Thus, a different algorithm is implemented in [Cold Diffusion Models](#). After executing a similar

noise-removal process, it adds a small amount of random noise back into the image, in other words, taking a step back. Following this, it attempts to correct the image by adjusting for this newly added noise in addition to the existing noise. This technique is more dynamic, where the process involves not just removal but also a intentional reintroduction of noise, followed by a final correction. The aim here is to potentially achieve a more accurate result. A UNet model is built and trained to implement this technique.

Algorithm 1 Standart Sampling¹

```

1: procedure STANDARTSAMPLING(degraded_image, max_timestep)
2:   for  $s \leftarrow \text{max\_timestep}$  down to 1 do
3:     estimated_original  $\leftarrow$  RESTORE(degraded_image,  $s$ )
4:     if  $s > 1$  then
5:       degraded_image  $\leftarrow$  DEGRADE(estimated_original,  $s - 1$ )
6:   return estimated_original

```

Algorithm 2 Improved Sampling for Cold Diffusion²

```

1: procedure IMPROVEDSAMPLING(degraded_image, max_timestep)
2:   for  $s \leftarrow \text{max\_timestep}$  down to 1 do
3:     estimated_original  $\leftarrow$  RESTORE(degraded_image,  $s$ )
4:     correction  $\leftarrow$  DEGRADE(estimated_original,  $s$ )
5:      $\quad - \text{DEGRADE}(\text{estimated\_original}, s - 1)$ 
6:     degraded_image  $\leftarrow$  degraded_image  $-$  correction
7:   return estimated_original

```

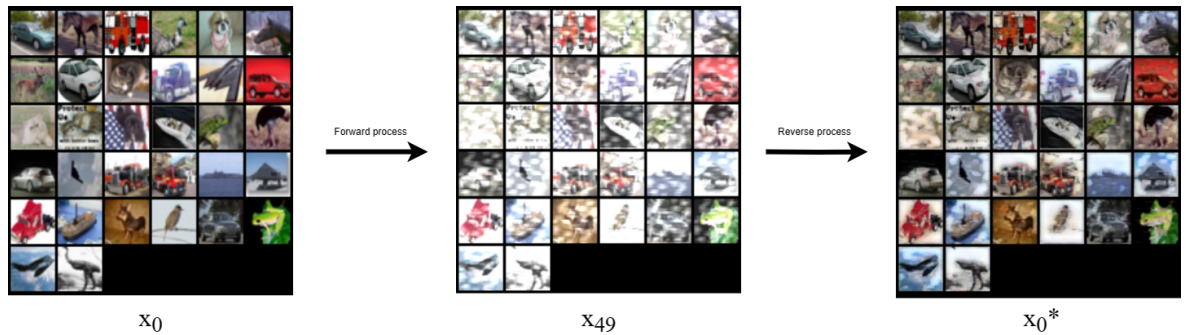


Figure 4.2. Forward and Reverse processes with snowification

¹Algorithm is adapted from Cold Diffusion paper [4].

²Algorithm is adapted from Cold Diffusion paper [4].

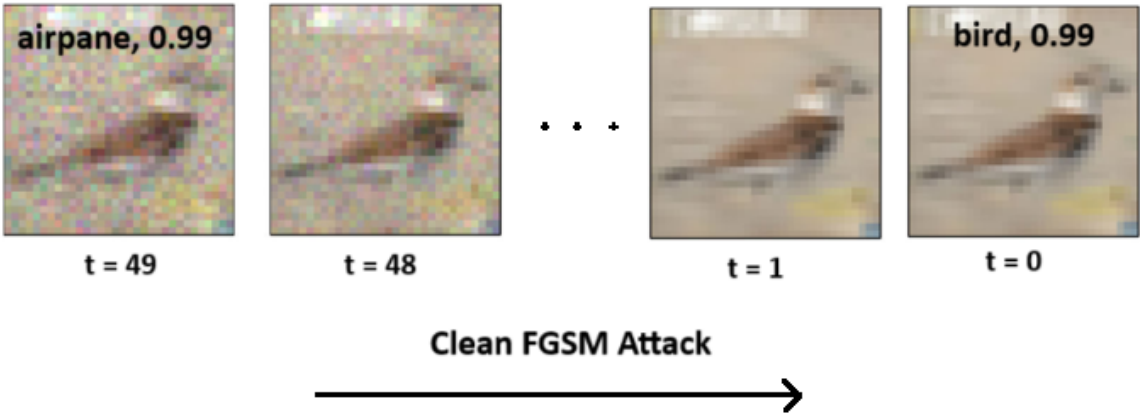


Figure 4.3. Reverse process of FGSM

4.3. Classification

We loaded training and test CIFAR-10 datasets and prepared for batch processing. The ResNet-18 model is adapted for ten classes of CIFAR-10. Only its final layer’s parameters are trained. Training occurred over three epochs. We saved the trained model to our Google Drive for later use. The model is then evaluated for accuracy on the test set. We test adversarial attacks through this classifier. Accuracy of the network on the test images: 93%

Classification examples:



Figure 4.4. ('cat',
0.9999656677246094)



Figure 4.5. ('deer',
0.9993529915809631)



Figure 4.6. ('airplane',
0.9063523411750793)



Figure 4.7. ('automobile',
0.9748400449752808)

4.4. Adversarial Attack

Adversarial attack is a type of corruption mechanism which aims to produce more deceptive results for model to predict in classification problem.

We implemented an adversarial attack on a pre-trained ResNet-18 model using the Fast Gradient Sign Method (FGSM) and assesses its impact on CIFAR-10 dataset classification.

To attack the images and make their classification wrong, we use Fast Gradient Sign Method (FGSM) as adversarial attack algorithm with the ResNet-18 model. It contains two phases.

- (i) **Classification Model Preparation:** Utilized a pre-trained ResNet-18 model, trained on ImageNet. We adapted for CIFAR-10 dataset with changing last layer and train a few epochs.
- (ii) **FGSM Algorithm:** The Fast Gradient Sign Method is an algorithm which generates adversarial examples. This technique uses the gradients of the neural network to create perturbations in the input images, making them 'adversarial'. It updates images as increasing its cost related to our ResNet-18. The formula used is:

$$\text{Adversarial Image} = \text{Original Image} + \epsilon \times \text{sign}(\nabla_{\text{image}} J(\theta, \text{image}, \text{label}))$$

where ϵ is a small perturbation (10/255 in our algorithm which is common practice).

The ResNet-18 model shows high accuracy on the original CIFAR-10 test set. but images which are generated using FGSM has significant drop in accuracy.

5. IMPLEMENTATION AND TESTING

5.1. Implementation

For implementation part we used department resources to eliminate lack of resources of GPU and RAM. We carry out our trainings that required complex calculations on this department computer through connecting remotely. Our codes are builded with a virtual machine in docker environment and pushed to Github Docker Registry. Then machine pulls this image and execute our codes. We reach this machine and its output via VPN connection.

We create and modify python scripts using different frameworks but mostly based on Tensorflow and PyTorch. Also we made collaborative coding and file sharing on Github and Google Drive.

In order to design encoder and decoder part, we utilized mostly "[Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise](#)" public repository. We created our custom repository according to requirements and methods. It consists different types of cold diffusion model forward implementation models, adversarial attack algorithms and some kind of related codes.

Repository has different kinds of UNet model implementation including newly published one. We used these models for decoder part after modify it to fit our experiment data.

Among forward process method we used snowification, blurring and custom FGSM implementation. We used mostly snowification during this process but our custom FGSM algorithm (mentioned in [Subsection 2.1.2. Adversarial Attack-based Degradation and Restoration](#)) is the most successful one to achieve higher accuracy after decoder step.

After training cold diffusion models, we try to establish adversarial attack and see the results. We implemented different algorithms such as Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) to obtain corrupted images for our trained model and we used these inputs to test with our cold diffusion models and compare them with original images also cleaned versions of them. In this experiments we couldn't get meaningful results with PGD algorithm. Considering also its complexity and execution time, using FGSM algorithm is the best choice for this purpose. But there is no necessity to apply adversarial attack when our forward process is also FGSM algorithm.

5.2. Testing

To make testing and evaluating the results, we used the predictions using the original images, attacked images, snowified images and cleaned images. For each of them overall accuracy is calculated. The difference between these predictions was also observed and it was inferenced that how effective the operations were. While difference between adversarial predictions and original predictions shows the success of adversarial attack, difference with cleaned image predictions shows how our model fix wrong predictions.

In addition to that, we saved images in particular time steps and merge them to create GIF image then visualize and present the changes in the manner of time steps.

6. RESULTS

6.1. Test results

6.1.1. Snowification-based

Prepared our training set with 50,000 CIFAR-10 samples, we trained the model in order to test our method.

While the test is running, snow is applied to these images. Afterwards, UNet tries to clear this snow. Images are saved after applying snow and after removing snow. So, at the end, we get 4 pairs of images for each image:

- (i) The original image
- (ii) The image with the adversarial attack
- (iii) The image with the snow added on the adversarial attack
- (iv) The image in which the snow added on the adversarial attack has been cleared.

- Set 50 timesteps for forward process.
- Trained for 100,000 epochs.

	10,000 normal	10,000 adversarial
Original	0.9306	0.5633
Snowified	0.5532	0.4985
Cleaned	0.884	0.6327

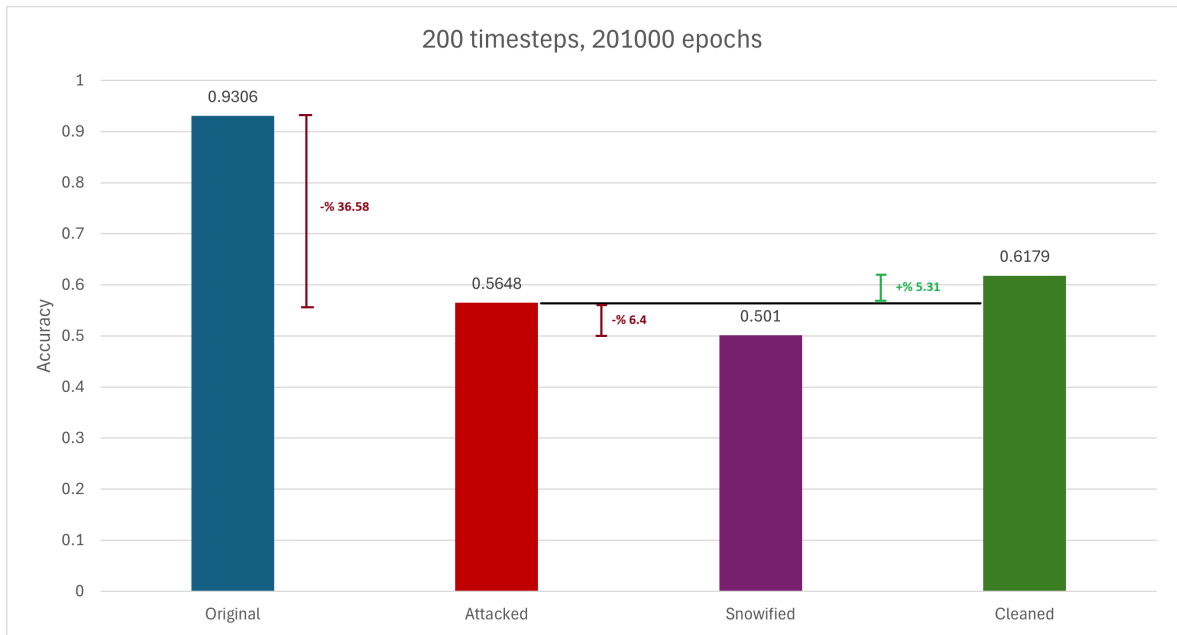


Figure 6.1. Snow test results 1

- Set 200 timesteps for forward process.
- Trained for 201,000 epochs.

	10,000 normal	9,567 adversarial
Original	0.9306	0.5648
Snowified	0.5532	0.501
Cleaned	0.884	0.6179

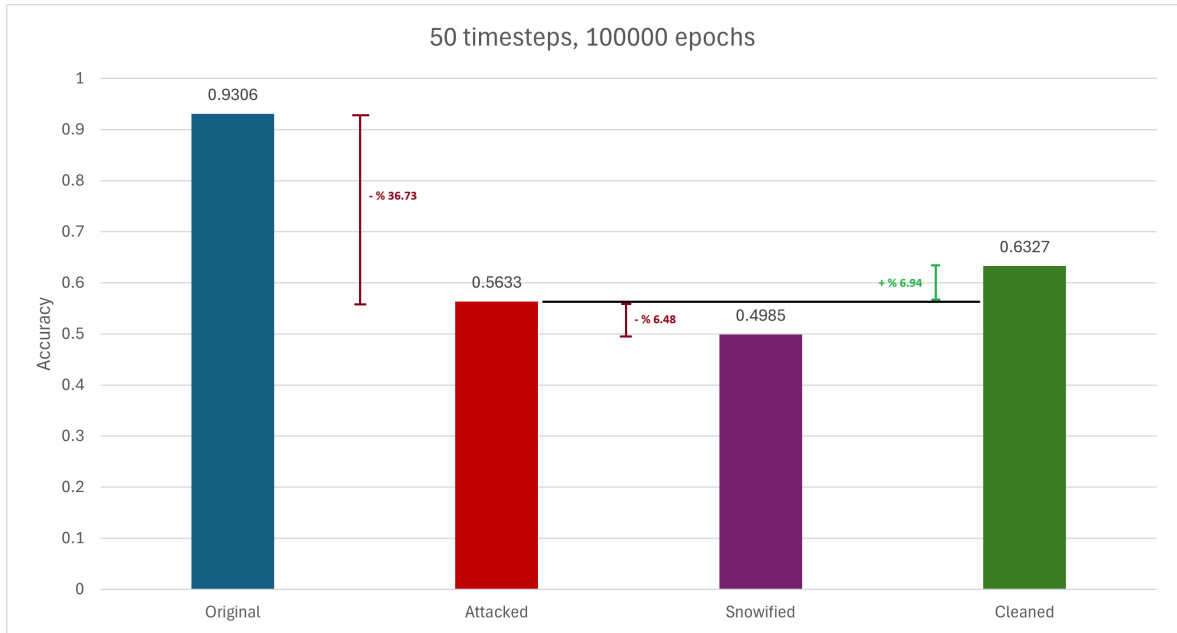


Figure 6.2. Snow test results 2

6.1.2. FGSM-based

We prepared our training set with 50,000 CIFAR-10 images and our test set with 10,000 CIFAR-10 images, and trained the model to test our method.

Minimum epsilon value is set to 0.01 (2.55/255), and maximum epsilon value is set to 0.3 (76.5/255). During testing, FGSM is initially applied with the highest epsilon value. Then with UNet, the attack is gradually cleaned using the [improved sampling](#).

Images are saved after applying attack and after removing attack. So, at the end, we get 3 images for each sample:

- (i) The original image
- (ii) The image with maximum adversarial attack
- (iii) Cleaned image after reverse process

- Minimum epsilon value is set to 0.01 (2.55/255)

- Maximum epsilon value is set to 0.3 (76.5/255)
- Set 50 timesteps for forward process.
- Trained for 67,000 epochs.

	10,000 normal
Original	0.9306
Attacked	0.3687
Cleaned	0.9171

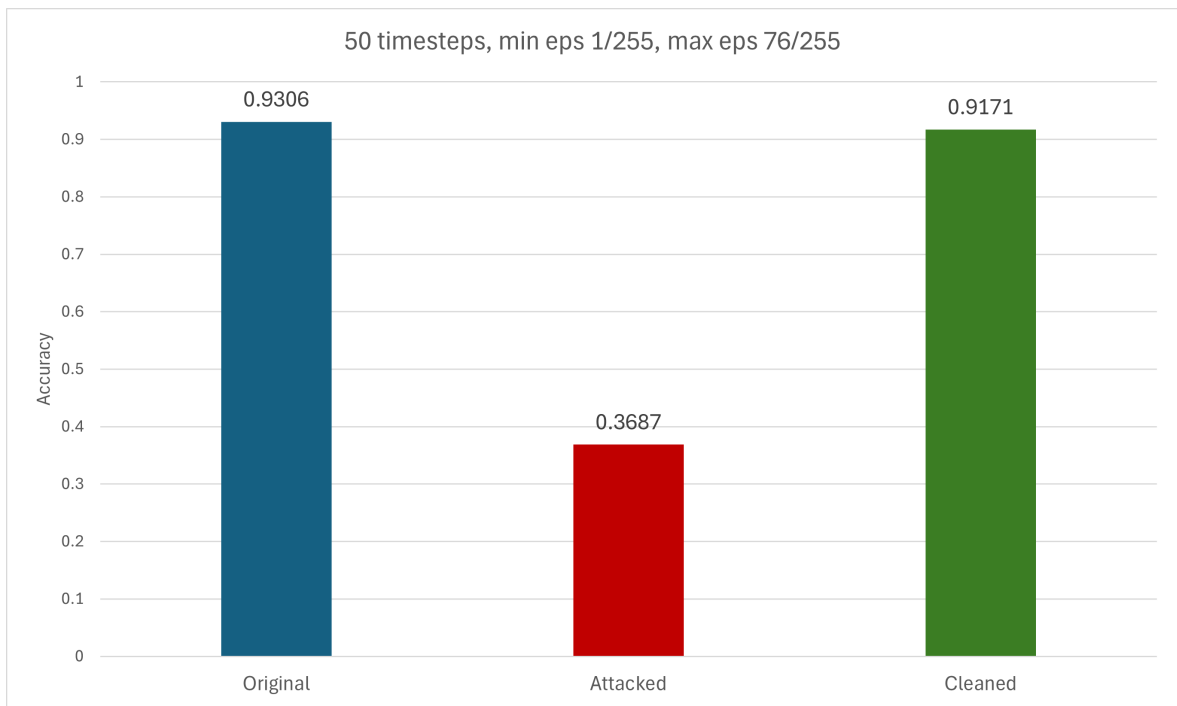


Figure 6.3. FGSM Test results

6.2. Interpretation of the results

The results indicate that the UNet model demonstrates capability in restoring images affected by both snowification and adversarial attacks. Key insights from the evaluation are as follows:

- **Baseline Performance:** The high classification accuracy of 93.06% on original

images establishes the baseline efficiency of the model on unaltered data.

- **Impact of Snowification:** Snowification significantly reduces classification accuracy to 55.32%. This degradation demonstrates the disruptive impact of the snow-like noise on the classifier’s performance.
- **Recovery from Snowification:** Restoring by the UNet model, accuracy recovers to 88.4%, demonstrating UNet’s effectiveness in reversing snowification.
- **Effectiveness Against Adversarial Attacks:** Adversarially attacked original images show a reduced accuracy to 56.33% from 93.06%. This reduction from the baseline highlights the vulnerability of the classifier to FGSM attacks.
- **Restoration from Adversarial Attacks:** The cleaned adversarially attacked images result in an improved accuracy of 63.27%. Although not as high as the baseline, this is a notable accuracy improvement from attack, which is 6.94%.
- **Extended Training and Adversarial Recovery:** Contrary to expectations, the second scenario with 200 timesteps and 201,000 epochs does not yield a better recovery rate. The accuracy after cleaning adversarially attacked images is 61.79%, which is lower than the recovery rate in the first scenario. This suggests that increased timesteps do not necessarily correspond to improved restoration from adversarial attacks.
- **Effectiveness Against Adversarial Attacks (FGSM-based):** The FGSM attack significantly decreases the classification accuracy to 36.87%. This drastic reduction highlights the severe impact of such adversarial perturbations on the classifier’s performance.
- **Restoration from Adversarial Attacks (FGSM-based):** After applying the UNet model for cleaning the adversarial noise, the classification accuracy improves to 91.71%. This recovery indicates a high effectiveness of the UNet model in mitigating the effects of adversarial attacks, closely approaching the baseline accuracy.

These findings highlight the UNet model’s potential in counteracting the effects of specific degradations. While the restoration from snowification is quite effective, the complete recovery from adversarial attacks in the FGSM-based experiment is highly

promising, showing a recovery rate that closely approaches the baseline accuracy. This suggests that the UNet model is capable of effectively mitigating adversarial perturbations, although further model refinement or alternative approaches may be necessary to enhance robustness against all types of adversarial attacks.

6.3. Possible Improvements

Models should also defend wider variety of adversarial attack methods to ensure comprehensive robustness. Additional adversarial techniques, such as PGD attack, for forward process could be developed.

Although our main focus was on the FGSM attack, we also experimented with the PGD attack using the snowification-based method. However, we obtained inconsistent and unsatisfactory results.

	10,000 normal
Attacked	0.002
Snowified	0.264
Cleaned	0.0923

Interestingly, when we add snow to the attacked images, the accuracy increases from 0.2% to 26%. When we clear this snow, the accuracy rate decreases again to 9%, which is not expected.

7. CONCLUSION

In this report, we explored the potential of using cold diffusion techniques to improve adversarial robustness in image classification models. We examined two primary approaches: snowification-based degradation and restoration, and direct adversarial attack-based degradation and restoration. Our methodology included detailed processes for forward and reverse operations, employing the UNet model for image restoration.

Our results demonstrated that the UNet model is effective in counteracting the effects of both snowification and adversarial attacks. Cold diffusion offers deterministic degradation, meaning specific changes can be made to images in a controlled way. These controlled degradations, like snowification, are applied on top of images that have already been attacked. When we reverse these controlled degradations, it helps to clear some of the effects of the attacks. In our study, we showed that this approach could restore lost accuracy to an extent. We also explored using the attack itself as the degradation method. By training the model to reverse the attack directly, we improved its ability to recognize and counteract adversarial perturbations. This method also successfully restored the accuracy to levels close to the original, demonstrating the effectiveness of our approach.

The findings highlight the importance of robust image processing techniques in enhancing the security and reliability of neural networks. While there is still room for improvement, especially in handling more complex adversarial attacks, the results are promising and suggest that further refinement of these techniques could result in more robust models.

In summary, our research underscores the potential of cold diffusion methods in defending against adversarial attacks, providing a foundation for future work in this area.

REFERENCES

1. Goodfellow, I., J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville and Y. Bengio, “Generative adversarial networks”, *Communications of the ACM*, Vol. 63, No. 11, pp. 139–144, 2020.
2. Ho, J., A. Jain and P. Abbeel, “Denoising Diffusion Probabilistic Models”, *Advances in Neural Information Processing Systems*, Vol. 33, pp. 6840–6851, 2020.
3. Silva, H. P., L. Seidenari and A. D. Bimbo, “DiffDefense: Defending against Adversarial Attacks via Diffusion Models”, arXiv preprint arXiv:2309.03702v1, 2023, <https://arxiv.org/abs/2309.03702>.
4. Bansal, A., E. Borgnia, H.-M. Chu, J. S. Li, H. Kazemi, F. Huang, M. Goldblum, J. Geiping and T. Goldstein, “Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise”, , 2022.
5. Hendrycks, D. and T. G. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations”, *International Conference on Learning Representations*, OpenReview.net, 2019.
6. Zhang, Y., J. He, L. Wei, D. Yang and C. Wang, “DENSEPURE: Understanding Diffusion Models for Adversarial Robustness”, *arXiv preprint arXiv:2210.02246*, 2022.
7. Xie, B., H. Xu, Z. Xiong, Y. Li and Z. Cai, “A Self-Supervised Purification Mechanism for Adversarial Samples”, *2022 IEEE International Conferences on Internet of Things (iThings) and IEEE Green Computing Communications (GreenCom) and IEEE Cyber, Physical Social Computing (CPSCoM) and IEEE Smart Data (SmartData) and IEEE Congress on Cybermatics (Cybermatics)*, pp. 501–509, 2022.